

Julia Ebert

Robotics Research & Development • Software Engineering • Boston, MA

🌐 juliaebert.com
✉ julia@juliaebert.com
🐙 github.com/jtebert
🌐 linkedin.com/in/jtebert

Skills

Tools	Linux • Docker • Git • ZeroMQ • Protobuf • ROS • PyTorch • Reinforcement learning • MuJoCo
Languages	C++ • Python • JavaScript • HTML/CSS
Everything else	Systems engineering • Computer-aided design (OnShape) • 3D printing • Laser cutting

Experience

Boston, MA 2026 –	Motional Staff Engineer, Autonomy › Contribute expertise, development, and code review to autonomous vehicle planning › Manage problem decomposition, testing, and metrics for new features in the AV planning stack. › Connect systems engineering and on-vehicle planning teams to ensure we develop the right solutions to the right problems.
Boston, MA 2023 – 2026	Fleet Robotics Autonomy Lead › Architect and implement the software stack from the ground up (hardware choice, communication protocols, Docker/CI, and UI development), enabling Fleet's first on-ship robot demonstrations. › Develop robust path planning (A*, RL) and execution systems for autonomous underwater navigation on complex surfaces, addressing challenges of limited sensing and harsh operating conditions. › Design and implement robot control systems, safety mechanisms, and a robust logging system. › Manage a team of 3 software engineers and supervise interns. › Lead software project planning and collaborate with stakeholders to manage timelines and risks.
Brighton, CO 2022 – 2023	Outrider Software Engineer, Mission Planning (Remote) › Spearheaded design and development (C++, ROS) of new multi-robot planning for Outrider's fleet of autonomous distribution yard trucks. › Led cross-functional project teams to create new robot behaviors toward commercial product goals. › Supported test site and customer deployments of the mission planning system.
Cambridge, MA 2016 – 2022	Harvard University Self-Organizing Systems Research Group , Prof. Radhika Nagpal Doctoral Researcher › Developed collective spatial decision-making algorithms for simulated and physical robot collectives. Includes bio-inspired and Bayesian decision and movement algorithms. › Collaborated with MIT Media Lab to create heterogeneous robot swarm for inspection on space stations, including algorithm development and hardware testing in microgravity. › Supervised undergraduate and masters research, and taught robotics and digital fabrication courses.

Education

Cambridge, MA 2022	Harvard University PhD in Computer Science › Thesis: <i>Distributed Decision-making for Inspection by Autonomous Robot Collectives</i> › Department of Energy Computation Science Graduate Fellow (DOE CSGF) • Siebel Scholar
London, UK 2016	Imperial College London Master of Research (MRes) in Bioengineering, with Distinction › Marshall Scholar • Thesis: <i>Assisting Balance Recovery with a Lower Limb Exoskeleton</i>
Boston, MA 2015	Northeastern University BS in Behavioral Neuroscience, Minor in Computer Science › Goldwater Scholar • German Academic Exchange Service (DAAD) Scholar • summa cum laude

Select Publications

J Meyers, M Schneider, **J Ebert** et al. (2025) SSAPy – Space Situational Awareness for Python. *JOSS*. [🔗](#)

J Ebert et al. (2022) A Hybrid PSO Algorithm for Multi-robot Target Search and Decision Awareness. *IROS 2022*. [🔗](#)

J Ebert et al. (2020) Bayes Bots: Collective Bayesian Decision-Making in Decentralized Robot Swarms. *ICRA 2020*. [🔗](#)

J Ebert et al. (2018) Multi-feature collective decision making in robot swarms. *AAMAS 2018*. [🔗](#)

S Bazzi, **J Ebert** et al. (2018) Stability and predictability in human control of complex objects. *Chaos*, 28, 10. . [🔗](#)